

Deep Learning – Graded Practice Exam

30 Questions — ~70 Marks

10 Easy + 12 Medium + 8 Hard — IITM End Term Pattern

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Difficulty	Questions	Marks	Target Time
EASY	10	~18	20 min
MEDIUM	12	~28	45 min
HARD	8	~24	55 min
Total	30	70	120 min

EASY (Q1–Q10)

EASY – Q1 [MCQ, 2 marks]

A sigmoid neuron outputs $\sigma(x) = \frac{1}{1+e^{-x}}$. What is $\sigma(0)$?
A. 0 B. 0.25 C. 0.5 D. 1

EASY – Q2 [MCQ, 2 marks]

A neural network has all weights initialized to zero, no biases, sigmoid activations everywhere including the output. What is the output for *any* input?
A. 0 B. 0.5 C. 1 D. Depends on input

EASY – Q3 [SA, 1 mark]

ReLU activation: $\text{ReLU}(x) = \max(0, x)$. Compute $\text{ReLU}(-5) + \text{ReLU}(3)$.

EASY – Q4 [MCQ, 2 marks]

Which of the following has learnable parameters?
A. Max Pooling layer B. Average Pooling layer C. Convolutional layer D. Dropout layer

EASY – Q5 [SA, 2 marks]

A 6×6 grayscale image is convolved with a single 3×3 filter, stride 1, no padding. What is the output size?

EASY – Q6 [MCQ, 2 marks]

In an LSTM, how many gate equations are there (including the candidate state)?

- A. 2 B. 3 C. 4 D. 5

EASY – Q7 [MCQ, 2 marks]

In masked self-attention with sequence length $T = 5$, what is the sum of all *diagonal* elements of the mask matrix?

- A. $-\infty$ B. 5 C. 0 D. -5∞

EASY – Q8 [SA, 1 mark]

Cross-entropy loss for a single sample: $L = -\ln(\hat{y}_{\text{true}})$. If $\hat{y}_{\text{true}} = 1.0$, what is L ?

EASY – Q9 [MCQ, 2 marks]

In a Transformer, which component has the keys and values from the encoder and queries from the decoder?

- A. Encoder self-attention B. Masked self-attention C. Cross attention D. FFN

EASY – Q10 [MCQ, 2 marks]

Which optimizer uses **both** momentum and adaptive learning rates?

- A. SGD with momentum B. AdaGrad C. RMSProp D. Adam

MEDIUM (Q11–Q22)**MEDIUM – Q11 [SA, 2 marks]**

A 7×7 RGB image ($D = 3$). We apply convolution with 8 filters of size 3×3 , stride 2, padding 1, **with bias**. How many learnable parameters?

MEDIUM – Q12 [SA, 2 marks]

Same padding formula. Image size $L \times L$, filter $F \times F$, stride = 1. What padding P is needed to keep output size = input size?

MEDIUM – Q13 [SA, 3 marks]

Skip-gram model, window size $d = 2$ (each side). Sentence: “Learning from data helps build intelligent systems for society” (9 words). How many unique (target, context) pairs?

MEDIUM – Q14 [SA, 3 marks]

CBOW model. Vocabulary = {apple, banana, cherry}. Context matrix:

$$C = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}$$

Word matrix:

$$W = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Compute $P(\text{banana}|\text{apple})$. (Context = “apple” alone, window = 1.)

MEDIUM – Q15 [SA, 2 marks]

RNN with $U = 2$, $W = 1$, $s_0 = 0$, sigmoid activation. Input $x_1 = 0$. Find s_1 .

MEDIUM – Q16 [MSQ, 2 marks]

Training error = 5%, test error = 48%. Which methods can reduce the test error? (Select all correct.)

A. L2 regularization **B.** Adding more layers **C.** Dropout **D.** Data augmentation **E.** NAG optimizer

MEDIUM – Q17 [SA, 2 marks]

LSTM with input dimension $n = 4$, hidden dimension $d = 3$, output dimension $k = 2$, with biases. How many total trainable parameters?

MEDIUM – Q18 [SA, 2 marks]

Transformer: $d_{\text{model}} = 128$, $h = 4$ heads, $d_v = 32$. Compute the number of elements in the attention output for a **single head** when sequence length $t = 20$.

MEDIUM – Q19 [SA, 2 marks]

FFN layer: $d_{\text{model}} = 128$, $d_{ff} = 512$, **with bias**. Total FFN parameters?

MEDIUM – Q20 [MCQ, 2 marks]

Step decay: $\eta_0 = 0.2$, halved every 100 *mini-batch updates*. Batch size = 50, dataset has 5000 samples. What is η at the end of epoch 1?

A. 0.2 **B.** 0.1 **C.** 0.05 **D.** 0.025

MEDIUM – Q21 [SA, 2 marks]

Given the following 3×3 input and 2×2 kernel, compute the convolution output (stride 1, no padding):

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

MEDIUM – Q22 [MCQ, 3 marks]

Match the following:

- | | |
|----------------|--|
| (P) Sigmoid | (1) Output: $(-1, 1)$, zero-centered |
| (Q) tanh | (2) Can cause dead neurons |
| (R) ReLU | (3) Output: $(0, 1)$, not zero-centered |
| (S) Leaky ReLU | (4) Avoids dead neuron problem |

- A. P→1, Q→3, R→2, S→4 B. P→3, Q→1, R→4, S→2
 C. P→3, Q→1, R→2, S→4 D. P→2, Q→1, R→3, S→4

HARD (Q23–Q30)**HARD – Q23 [SA, 3 marks]**

Attention mechanism: $\text{score}(s_{t-1}, h_j) = V_{\text{att}}^T \tanh(U_{\text{att}} \cdot s_{t-1} + W_{\text{att}} \cdot h_j)$.
 Given: $h_1 = [1, -1]^T$, $h_2 = [0, 2]^T$, $s_{t-1} = [1, 1]^T$, $U_{\text{att}} = W_{\text{att}} = I$ (identity), $V_{\text{att}} = [1, 1]^T$.
 Compute the attention weights α_1 and α_2 .

HARD – Q24 [SA, 3 marks]

Transformer: $t = 16$, $d_{\text{model}} = 256$, $h = 8$, $d_v = 32$. How many parameters in the Multi-Head Attention (no bias)?

HARD – Q25 [SA, 3 marks]

Co-occurrence matrix. Corpus (stop words removed):

1. “students read books library”
2. “teachers read books class”
3. “students teachers discuss library”

Window = 1. Compute $X[\text{read}, \text{books}]$ and determine the sign of PMI(read, books).

HARD – Q26 [SA, 3 marks]

CBOW gradient update. After forward pass, the predicted probability for correct word w is $\hat{y}_w = 0.3$. Context embedding $u_c = [2, -1]^T$. Learning rate $\eta = 0.1$. Current word vector $v_w = [1, 1]^T$.

Compute the updated v_w^{new} using the gradient rule: $v_w^{\text{new}} = v_w - \eta \cdot u_c(\hat{y}_w - 1)$.

HARD – Q27 [SA, 3 marks]

ResNet with two residual blocks. $f(x) = x + \text{ReLU}(w_1 \cdot x)$ for block 1, $f(x) = x + \text{ReLU}(w_2 \cdot x)$ for block 2. Given $w_1 = 2$, $w_2 = -3$, input $x = 1$.

Compute the final output after both blocks.

HARD – Q28 [SA, 3 marks]

LSTM. Input $x_t \in \mathbb{R}^5$, hidden $h_t \in \mathbb{R}^6$, output $\hat{y}_t \in \mathbb{R}^4$, sequence length $T = 10$, **with bias**. Total trainable parameters?

HARD – Q29 [SA, 3 marks]

Full CNN pipeline. Input: $8 \times 8 \times 3$ image.

Layer 1: Conv(3×3 , 16 filters, stride 1, padding 1, bias)

Layer 2: MaxPool(2×2 , stride 2)

Layer 3: Conv(3×3 , 32 filters, stride 1, padding 0, bias)

Compute: (a) output size after each layer, (b) total learnable parameters.

HARD – Q30 [SA, 3 marks]

Attention scores: batch size $B = 8$, sequence length $T = 64$, heads $h = 4$. How many total attention scores are computed? If each score is a 32-bit float, how many bytes of memory?

SOLUTIONS

EASY

Q1: C. $\sigma(0) = \frac{1}{1+1} = 0.5$

Q2: B. All pre-activations = 0 $\rightarrow \sigma(0) = 0.5$ at every layer $\rightarrow$ output = 0.5.

Q3: $\text{ReLU}(-5) + \text{ReLU}(3) = 0 + 3 = \mathbf{3}$

Q4: C. Convolutional layer (has kernel weights and optionally bias).

Q5: $W_{\text{out}} = 1 + \frac{6-3}{1} = 4$. Output: 4×4 .

Q6: C. 4 equations: forget gate, input gate, output gate, candidate state $\tilde{s}_t$.

Q7: C. Diagonal = 0 (tokens can attend to themselves). Sum = 0.

Q8: $L = -\ln(1) = \mathbf{0}$ (perfect prediction has zero loss).

Q9: C. Cross attention.

Q10: D. Adam (first moment = momentum, second moment = adaptive LR).

MEDIUM

Q11: Output size: $1 + \frac{7-3+2}{2} = 1 + 3 = 4$. Output: $4 \times 4 \times 8$.

Params: $F \times F \times D_{\text{in}} \times K + K = 3 \times 3 \times 3 \times 8 + 8 = 216 + 8 = \mathbf{224}$.

Q12: $L = 1 + \frac{L-F+2P}{1} \Rightarrow 0 = -F + 2P \Rightarrow P = \frac{F-1}{2}$ (for odd F).

Q13: 9 words, $d = 2$. Count per word:

Word index	Pairs
0	$\min(2, 0) + \min(2, 8) = 0 + 2 = 2$
1	$1 + 2 = 3$
2	$2 + 2 = 4$
3	$2 + 2 = 4$
4	$2 + 2 = 4$
5	$2 + 2 = 4$
6	$2 + 2 = 4$
7	$2 + 1 = 3$
8	$2 + 0 = 2$

Total = $2 + 3 + 4 + 4 + 4 + 4 + 4 + 3 + 2 = \mathbf{30}$.

Q14: Context = “apple” (index 0). $u_{\text{apple}} = C[:, 0] = [1, 2]^T$.

$$\text{Scores: } z = W^T u = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Wait, W is 3×2 , so W^T is 2×3 . We need $z = W \cdot u$ where $W \in \mathbb{R}^{3 \times 2}$:

$$z = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$$

Softmax: $e^{-1} = 0.368$, $e^2 = 7.389$, $e^{-1} = 0.368$. Sum = 8.125.

$P(\text{banana}|\text{apple}) = 7.389/8.125 \approx \mathbf{0.909}$.

Q15: $s_1 = \sigma(2 \cdot 0 + 1 \cdot 0) = \sigma(0) = \mathbf{0.5}$.

Q16: A, C, D. Low train error + high test error = overfitting. L2 reg, dropout, data augmentation fix it. More layers makes it worse. NAG is an optimizer (doesn't fix overfitting).

Q17: $4d(n + d + 1) + k(d + 1) = 4 \cdot 3(4 + 3 + 1) + 2(3 + 1) = 12 \cdot 8 + 8 = 96 + 8 = \mathbf{104}$.

Q18: Output per head = $t \times d_v = 20 \times 32 = \mathbf{640}$ elements.

Q19: $2 \cdot 128 \cdot 512 + 512 + 128 = 131072 + 512 + 128 = \mathbf{131712}$.

Q20: Updates per epoch = $5000/50 = 100$. After epoch 1: 100 updates. Decay at 100 updates $\Rightarrow \eta = 0.2 \times 0.5 = \mathbf{0.1}$. **Answer: B.**

Q21: Output size = $1 + (3 - 2)/1 = 2 \times 2$.

$(0, 0): 1 \cdot 1 + 2 \cdot 0 + 4 \cdot 0 + 5 \cdot 1 = 6$

$(0, 1): 2 \cdot 1 + 3 \cdot 0 + 5 \cdot 0 + 6 \cdot 1 = 8$

$(1, 0): 4 \cdot 1 + 5 \cdot 0 + 7 \cdot 0 + 8 \cdot 1 = 12$

$(1, 1): 5 \cdot 1 + 6 \cdot 0 + 8 \cdot 0 + 9 \cdot 1 = 14$

Output = $\begin{bmatrix} 6 & 8 \\ 12 & 14 \end{bmatrix}$

Q22: C. P $\rightarrow$ 3 (sigmoid: (0,1), not zero-centered), Q $\rightarrow$ 1 (tanh: (-1,1), zero-centered), R $\rightarrow$ 2 (ReLU: dead neurons), S $\rightarrow$ 4 (Leaky ReLU: no dead neurons).

HARD

Q23:

Score for h_1 : $Us_{t-1} + Wh_1 = [1, 1] + [1, -1] = [2, 0]$. $\tanh([2, 0]) = [0.964, 0]$.

$e_1 = V^T \tanh = [1, 1][0.964, 0]^T = 0.964$.

Score for h_2 : $[1, 1] + [0, 2] = [1, 3]$. $\tanh([1, 3]) = [0.762, 0.995]$.

$e_2 = [1, 1][0.762, 0.995]^T = 1.757$.

Softmax: $\exp(0.964) = 2.624$, $\exp(1.757) = 5.795$. Sum = 8.419.

$\alpha_1 = 2.624/8.419 \approx \mathbf{0.312}$, $\alpha_2 = 5.795/8.419 \approx \mathbf{0.688}$.

Q24: $d_k = d_q = 256/8 = 32$, $d_v = 32$.

Per head: $32 \cdot 256 + 32 \cdot 256 + 32 \cdot 256 = 3 \times 8192 = 24576$.

8 heads: $24576 \times 8 = 196608$.

W_O : $256 \times (8 \times 32) = 256 \times 256 = 65536$.

Total = $196608 + 65536 = \mathbf{262144}$.

Q25: Sentences after removing stops:

1: students-read, read-books, books-library

2: teachers-read, read-books, books-class

3: students-teachers, teachers-discuss, discuss-library

$X[\text{read, books}] = 2$ (appears in sentences 1 and 2).

For PMI: total pairs $N = \text{sum of all entries} = 2(3 + 3 + 3) = 18$ (symmetric matrix).

$\text{count}(\text{read}) = 2 + 2 = 4$ (row sum), $\text{count}(\text{books}) = 2 + 1 + 1 = 4$ (row sum).

$\text{PMI} = \ln \frac{2 \times 18}{4 \times 4} = \ln \frac{36}{16} = \ln 2.25 \approx 0.811 > 0$. **Positive.**

Q26: $v_w^{\text{new}} = v_w - \eta \cdot u_c(\hat{y}_w - 1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} (0.3 - 1)$
 $= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} (-0.7) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.14 \\ -0.07 \end{bmatrix} = \begin{bmatrix} 1.14 \\ 0.93 \end{bmatrix}$

Q27:

Block 1: $f(1) = 1 + \text{ReLU}(2 \times 1) = 1 + 2 = 3$.

Block 2: $f(3) = 3 + \text{ReLU}(-3 \times 3) = 3 + \text{ReLU}(-9) = 3 + 0 = \mathbf{3}$.

Q28: $n = 5$, $d = 6$, $k = 4$.

$4d(n + d + 1) + k(d + 1) = 4 \cdot 6(5 + 6 + 1) + 4(6 + 1) = 24 \cdot 12 + 28 = 288 + 28 = \mathbf{316}$.

$T = 10$ doesn't matter (parameters shared across time).

Q29:

(a) Layer 1: $1 + \frac{8-3+2}{1} = 8$. Output: $8 \times 8 \times 16$.

Layer 2 (MaxPool): $1 + \frac{8-2}{2} = 4$. Output: $4 \times 4 \times 16$.

Layer 3: $1 + \frac{4-3+0}{1} = 2$. Output: $2 \times 2 \times 32$.

(b) Layer 1: $3 \times 3 \times 3 \times 16 + 16 = 432 + 16 = 448$.

Layer 2: 0 (pooling).

Layer 3: $3 \times 3 \times 16 \times 32 + 32 = 4608 + 32 = 4640$.

Total = $448 + 0 + 4640 = \mathbf{5088}$.

Q30: Total scores = $B \times h \times T \times T = 8 \times 4 \times 64 \times 64 = \mathbf{131072}$.

Memory: $131072 \times 4 = 524288$ bytes = **512** KB.